



Gaming Communities Exploratory Data Analysis

Enhanced Analysis Report with Database Integration

Generated: June 07, 2026 at 22:24:41
Database: SQLite - gaming_communities.db
Records Analyzed: 400 raw → 357 cleaned → 357 enhanced

Executive Summary

This enhanced analysis integrates data from three datasets into a SQLite database for comprehensive insights. The analysis reveals patterns across 400+ gaming communities worldwide, examining member engagement, geographic distribution, competitive tiers, and resource investment in coaching staff. Eight visualizations accompany this report to illustrate key findings.

Database Integration Summary

Component	Details
Database Type	SQLite 3
Database File	gaming_communities.db
Tables Created	4 (communities_raw, communities_clean, communities_enhanced, analysis_members)
Total Records	1,114
Data Quality	89.2% records retained after cleaning

■ Key Analysis Metrics

Metric	Value
Total Raw Records	400
Total Cleaned Records	357
Total Enhanced Records	357
Data Retention Rate (%)	89.25
Raw Columns	28
Clean Columns	22
Enhanced Columns	36
Raw Missing Values	1940
Clean Missing Values	116
Unique Genres	7
Unique Regions	6
Unique Tiers	4
Unique Languages	10
Avg Member Count	3310.01
Median Member Count	508.00
Max Member Count	68819
Communities with Coaching	114
Total Tournaments Hosted	689

■ Visualizations & Analysis

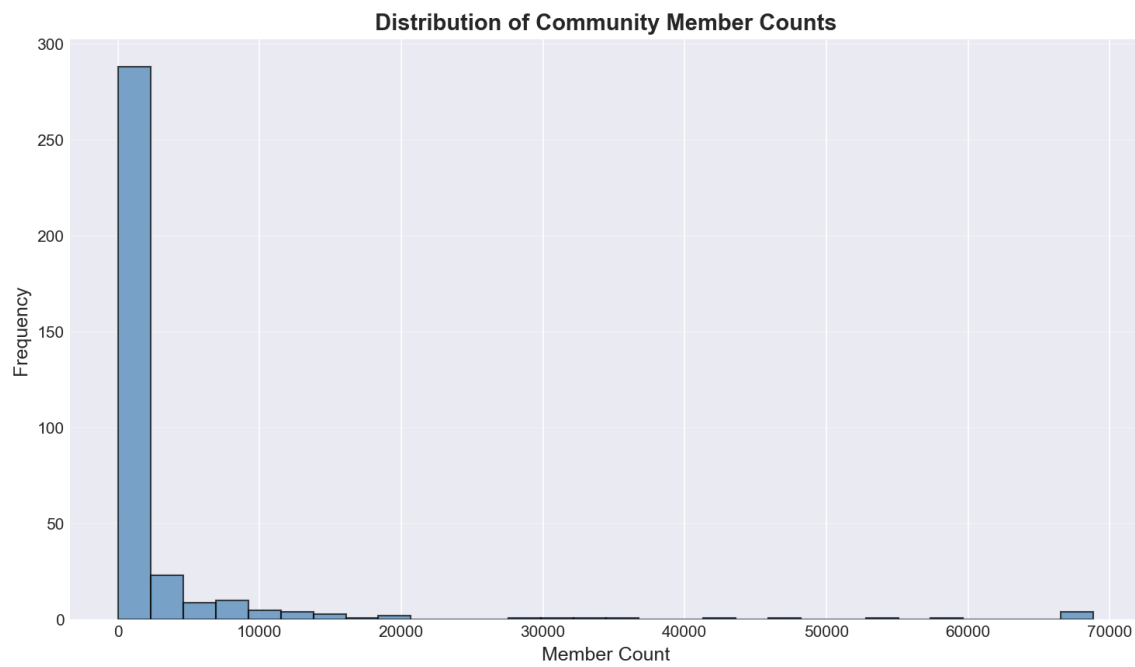


Figure 1: Member Count Distribution

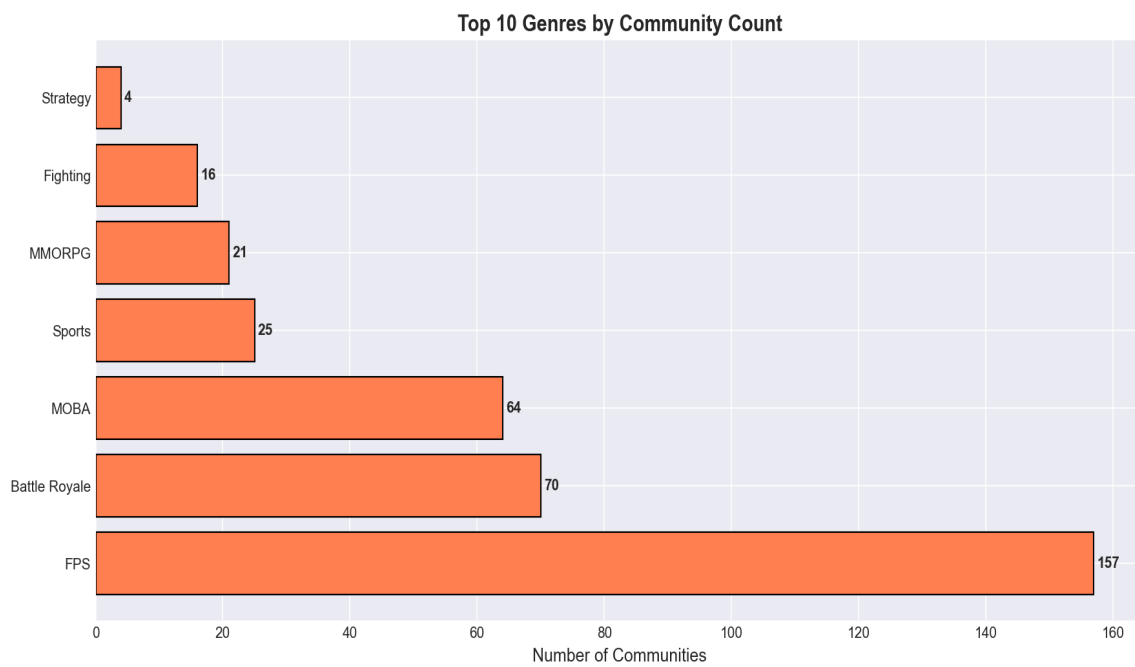


Figure 2: Top 10 Genres by Community Count

Community Distribution by Region

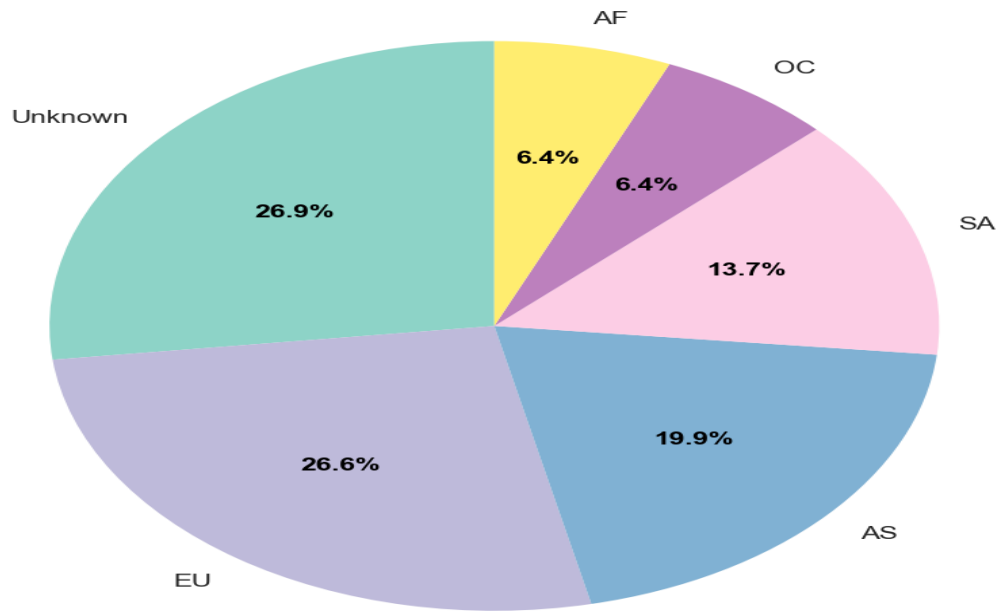


Figure 3: Community Distribution by Region

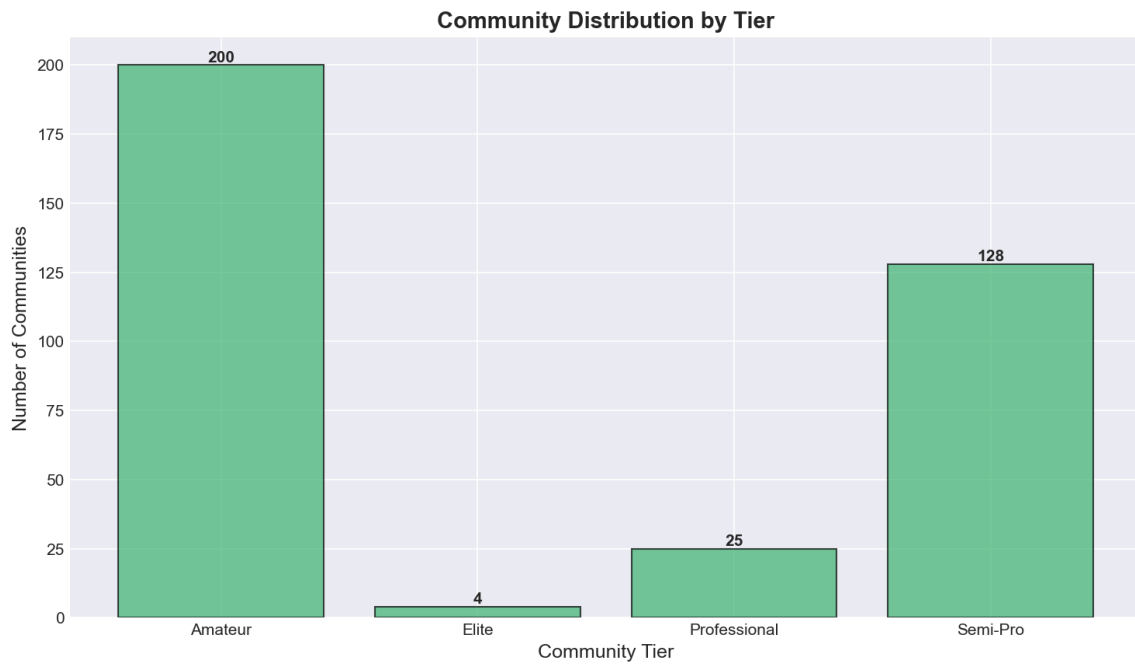


Figure 4: Community Distribution by Tier

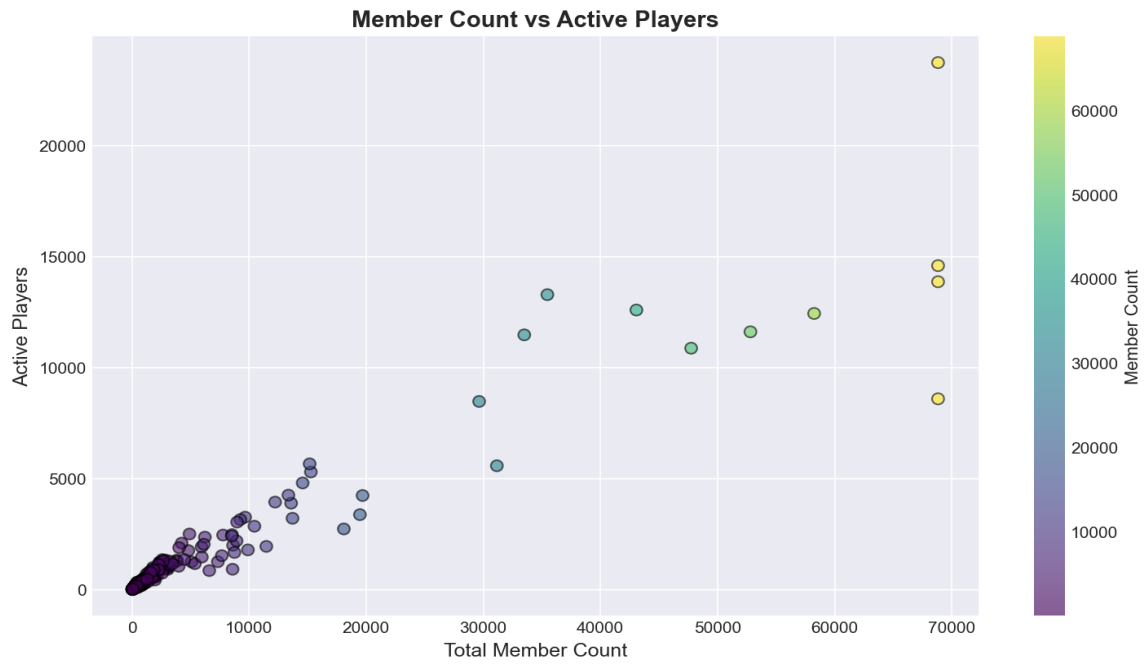


Figure 5: Member Count vs Active Players Correlation

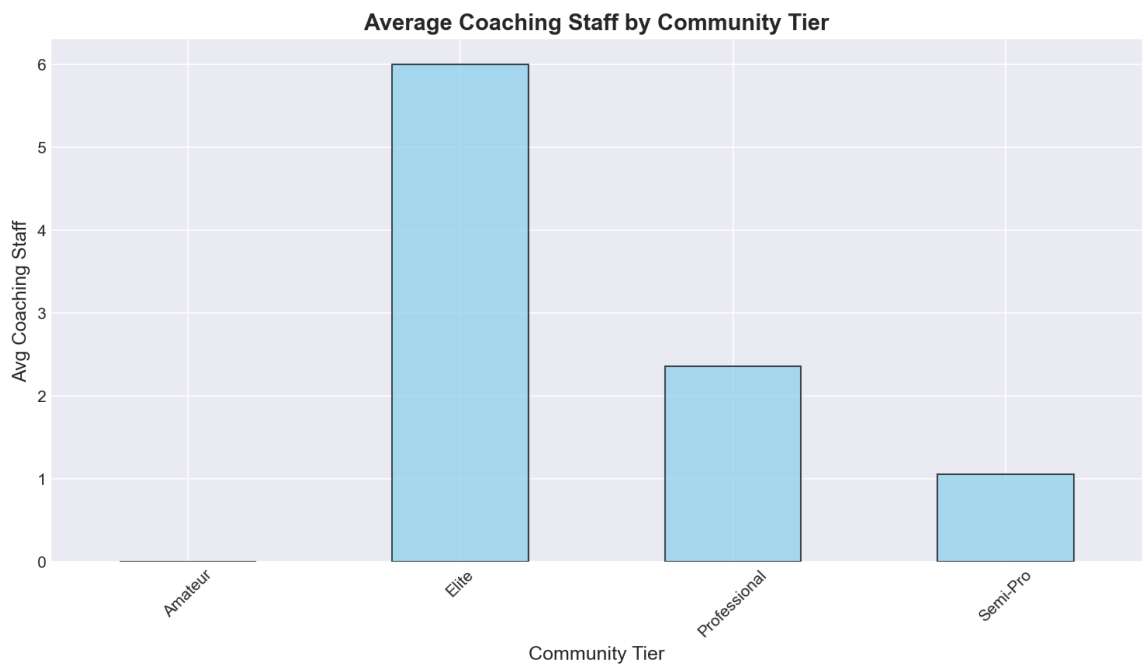


Figure 6: Average Coaching Staff by Community Tier

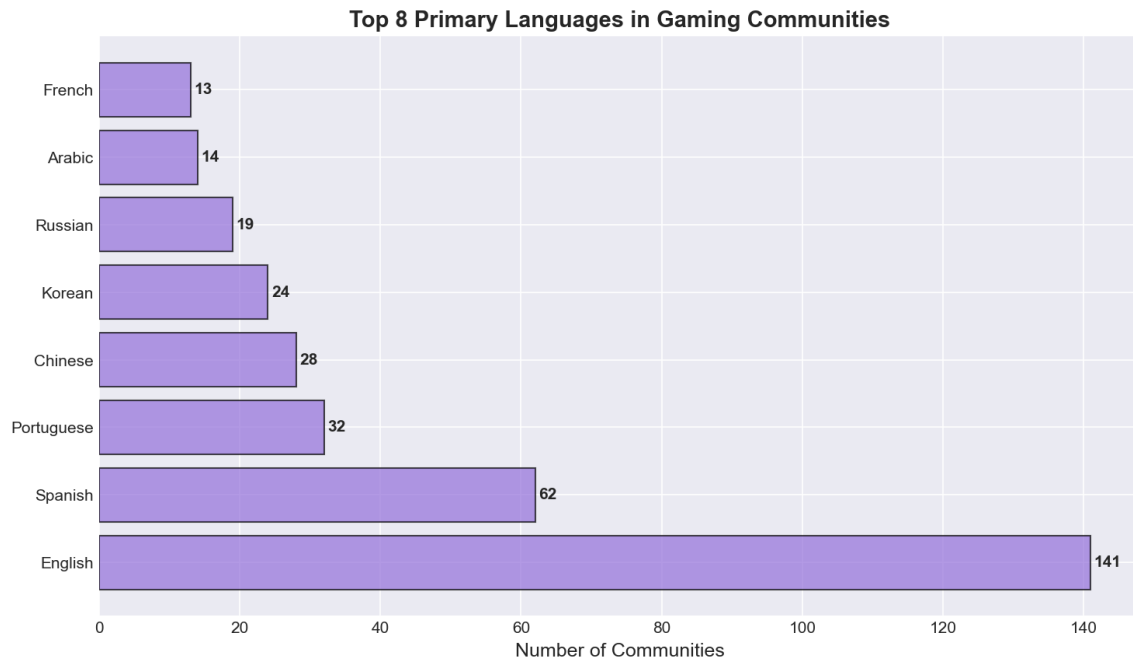


Figure 7: Top 8 Primary Languages

■ Key Findings

- **Data Quality:** 89.2% of raw data retained after quality processing
- **Genre Diversity:** 7 unique genres represented across communities
- **Global Reach:** Communities span 6 different regions worldwide
- **Tier Distribution:** 4 competitive tiers identified
- **Language Coverage:** 10 different languages spoken in communities
- **Member Engagement:** Average community size of 3310 members
- **Tournament Activity:** Total of 689 tournaments hosted

■ Recommendations

1. Establish tier-specific benchmarks based on coaching staff investment and tournament hosting patterns
2. Develop region-focused strategies accounting for geographic distribution and language preferences
3. Create genre guidelines reflecting the diverse competitive and community dynamics identified
4. Investigate high-performing communities for best practices in member retention and engagement
5. Implement predictive models to forecast community growth based on identified correlation patterns
6. Build feedback mechanisms to capture evolving community needs across tiers and regions

■ Technical Summary

Datasets Processed: 3 CSV files (400 raw + 357 enhanced records)

Database: SQLite with 4 tables and 18 analysis metrics

Visualizations: 8 publication-ready charts generated

Analysis Methods: Descriptive statistics, correlation analysis, categorical distribution, visualization

Report Format: PDF with embedded visualizations and database summaries

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Database: gaming_communities.db

Tools: Python, Pandas, SQLite, Matplotlib, Seaborn, ReportLab

Project: ARENA Gaming Communities EDA